

# **Gujarat Council on Science & Technology**

Dept. of Science & Technology, Govt. of Gujarat

## **Technical Report for Level-2 Proof of Concept (PoC) Stage for ROBOFEST-GUJARAT 5.0 Competition**

Level: Senior

Category: Aerial Robotics: Minefield Navigation Challenge

Application No: R520250962

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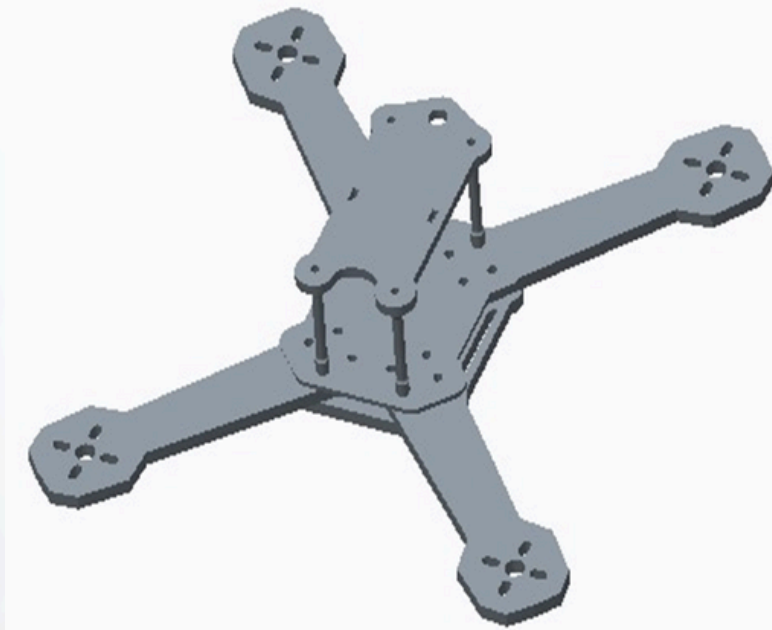
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## **Project S.C.A.N. (Swarm Coordinated Autonomous Navigation)**

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# Perspective view of Robot and Description



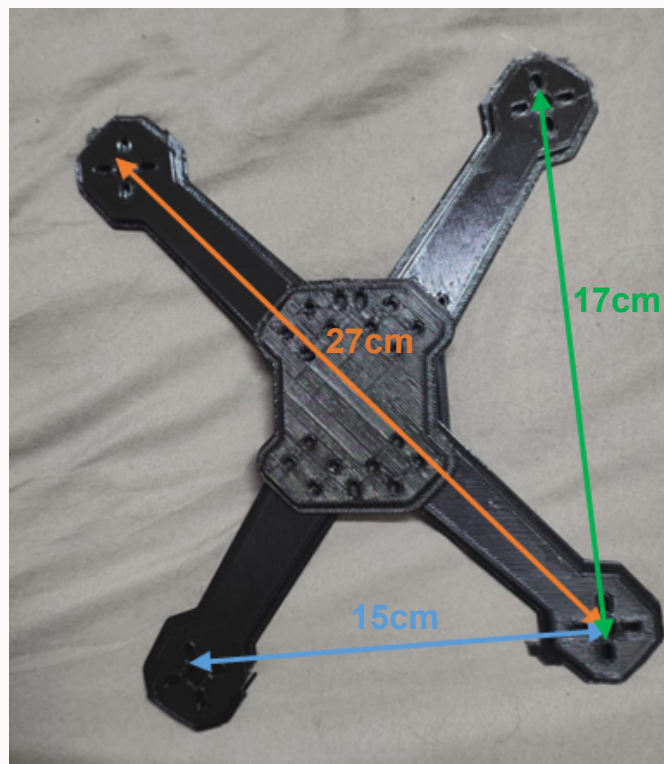
## **LIGHTWEIGHT AUTONOMOUS DRONE FOR MINEFIELD NAVIGATION AND HAZARD DETECTION.**

Powered by an ESP32-S3 flight controller with MPU6050 stabilization and cascaded PID control, it ensures smooth, accurate flight. An ESP32-CAM with Edgelpulse models enables onboard visual mine detection, while decentralized swarm architecture supports multi-drone coordination without a ground station. Built with Foxeer Datura 1404 motors, SimonK ESCs, and a custom 3D-printed frame, the platform delivers low-cost, expandable, and reliable field performance

# Logical Steps in the Working of Robot

## A. Design Outlay of the Robot

- A custom lightweight X-frame quadrotor (<454g) optimized for stability, agility, and Centre of Gravity(CoG) balance.
- Powered by brushless DC motors with 3016 propellers, ESP32-S3 flight control, and cascaded PID stabilization.
- ESP32-CAM enables onboard mine detection, while swarm coordination ensures reliable, low-cost field performance.



- Mechanical Specifications:(Approx)
  - Diagonal Length (Motor-to-Motor): 27 cm
  - Length (Motor-to-Motor): 17cm
  - Width (Motor-to-Motor): 15 cm
  - Height: 15 cm



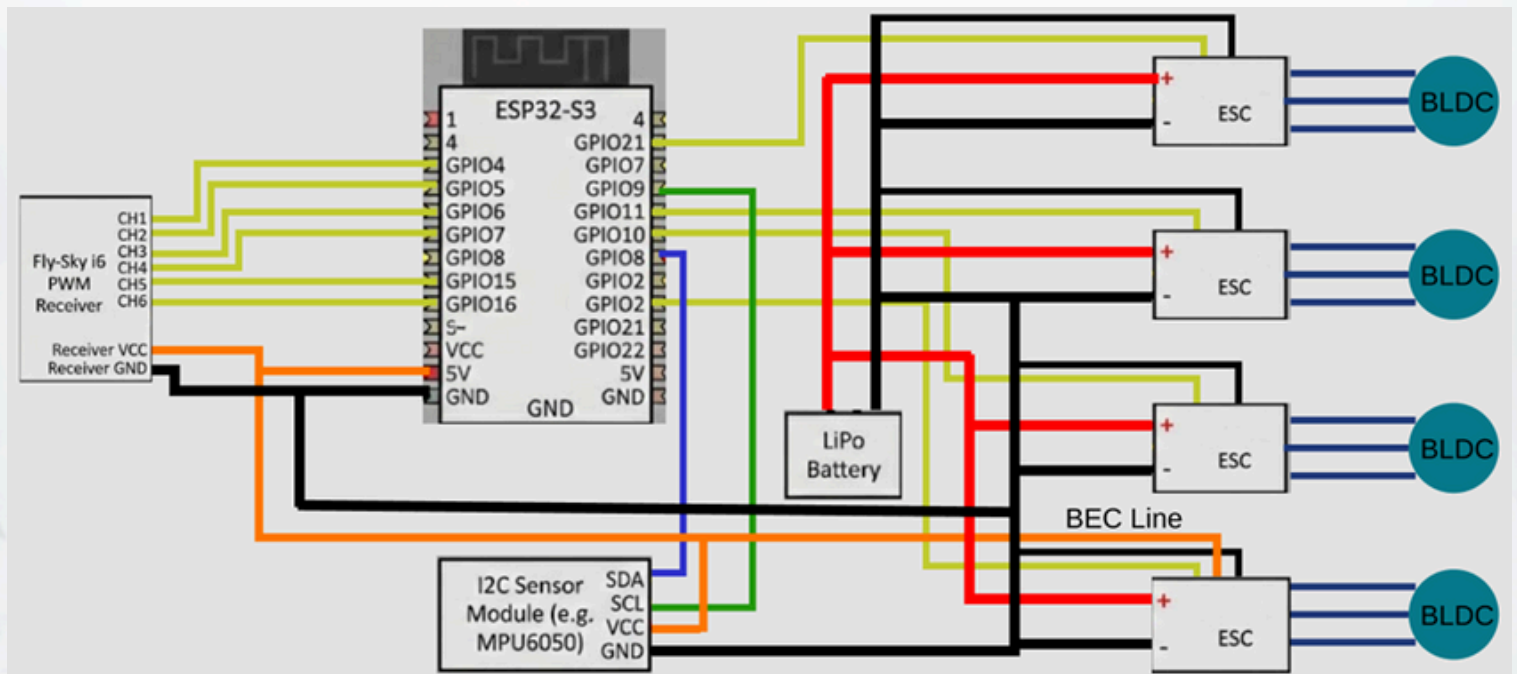
# Logical Steps in the Working of Robot

## B. Drive System and Controller

The drone is operated via a Fly-Sky i6 transmitter and PWM receiver. Channel signals for throttle, roll, pitch, yaw, and auxiliaries are read by the ESP32 through interrupt-driven pins and used as references for cascaded PID controllers. The ESP32 outputs PWM to the ESCs, ensuring stable motor actuation and reliable manual control alongside onboard autonomy and safety logic.

## C. Electronics & Instrumentation

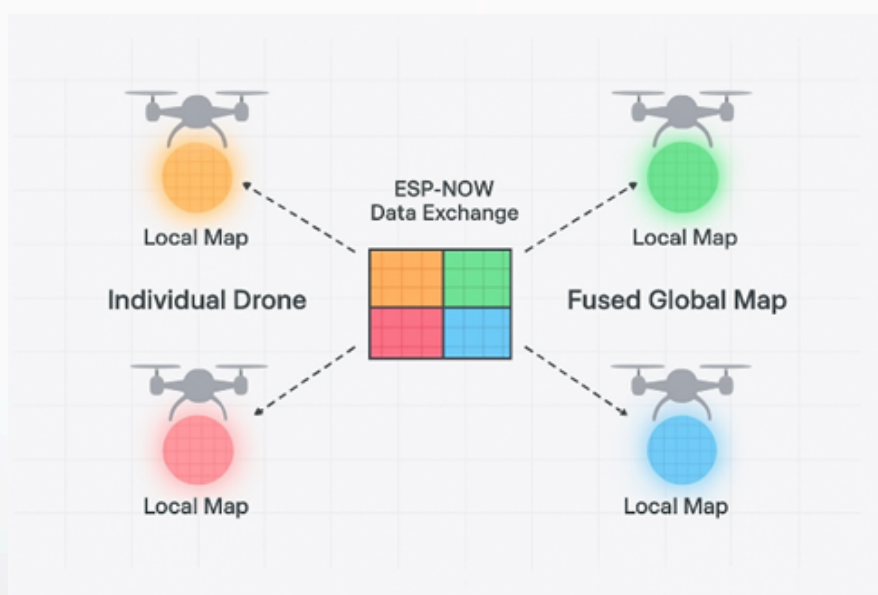
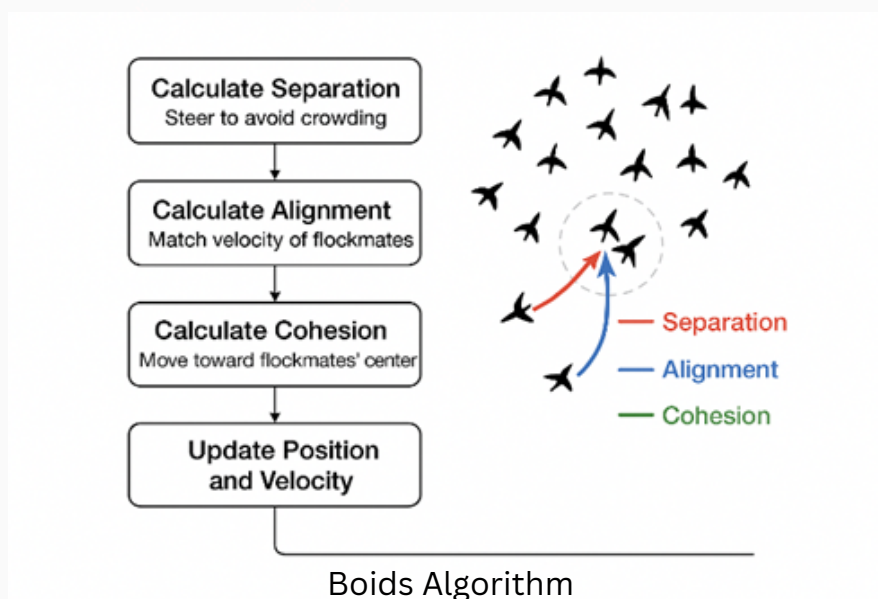
The avionics stack integrates an MPU6050 IMU for attitude sensing, with an ESP32 as the main controller and an ESP32-CAM running Edgelpulse models for mine detection. Propulsion uses Foxeer Datura 1404–3850KV motors driven by SimonK 30A ESCs, while a BMP280 barometric sensor supports altitude estimation. Together, these modules form a compact and efficient flight and perception suite



# Logical Steps in the Working of Robot

## D. Working Level Software

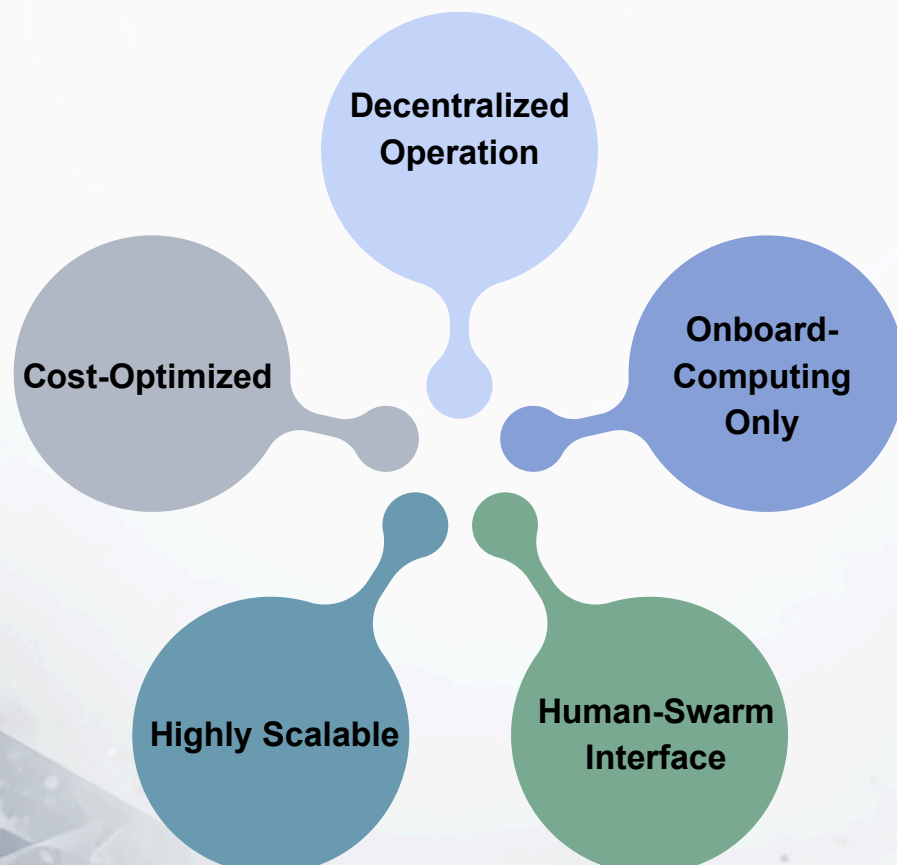
The ESP32 executes a cascaded PID controller using RC inputs and MPU6050 data for stable attitude control. A complementary filter supports Angle and Rate PID loops driving the ESCs. Higher-level autonomy integrates Project S.C.A.N. for swarm coordination, C-SLAM for mapping, and ESP32-CAM for mine detection, with BMP280 feedback planned for altitude estimation.



C-Slam Implementation

# Features of the Design

- 1) Fully Decentralized Operation:** The swarm functions without a central ground station and remains operational even if individual drones fail.
- 2) Onboard-Only Computing:** All core processes—including SLAM, pathfinding, stabilization, and AI inference—run locally on the ESP32-S3 modules.
- 3) Multi-Modal Human–Swarm Interface:** The system supports both gesture-based control and voice-command interpretation for intuitive operator interaction.
- 4) Highly Scalable Architecture:** Additional drones can be integrated into the swarm without modifying the existing software framework.
- 5) Cost-Optimized Design:** Relies on readily available off-the-shelf components such as ESP32 processors and standard brushless motors, enabling an affordable yet capable solution.



## (Electronics and Mechanical)

### A. Controller of the Robot

## 1. Electrical Assembly & Wiring

- a. 4× SimonK 30A Electronic Speed Controllers (ESCs)
- b. Motor power distribution wiring (XT60 connectors, silicon wires)
- c. Jumper wires and signal harnesses
- d. Common ground bus for ESC–Controller interfacing
- e. Battery-to-ESC power leads

## 2. Electronics & Instrumentation Assembly

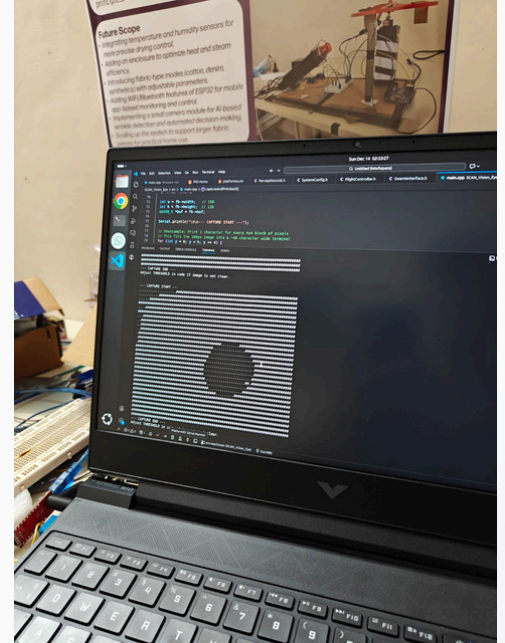
- MPU-6050 IMU module for stabilization
- ESP32-CAM module for mine detection
- FlySky iA6 Receiver for RC control input
- Onboard 5V/3.3V voltage regulators

### 3. Electronics Components Used in Fabrication

- Soldering station and solder wire
- 3D Printer (for frame, cage, sensor mounts, electronic holders)
- Laptop/PC for firmware development, flashing, and calibration
- Heat-shrink tubing for electrical insulation

#### 4. Tools & Equipment Used

- a. Power drill (mounting holes, frame modifications)
- b. Rotary grinder(custom motor fitting, trimming)
- c. Metal/wood file set (edge finishing, alignment)
- d. Digital multimeter (continuity checks, voltage validation)
- e. Hot glue gun / epoxy for structural reinforcement



## Mine Detection



## FlySky I6-Receiver



## Electronic Speed Controller(ESC)



## ESP Camera Module



# **Listing of all the hardware / component / equipment**

## **(Electronics and Mechanical)**

### **B. Mechanical Components**

#### **1. Mechanical Modules**

- a. 4× Foxeer Datura 1404 3850KV Brushless Motors
- b. GEMFAN 3016 tri-blade propellers
- c. Battery strap / Hook-and-loop fasteners

#### **2. Fabrication Components**

- a. Custom-designed chassis and protective cage (3D-printed or composite)
- b. Motor mounting arms/plates
- c. Fasteners (M2/M3 screws, lock nuts, spacers, standoffs)
- d. Sensor mounts (camera cage, barometer housing)

#### **3. Tools and Equipment Used**

- a. Screwdrivers
- b. Wire cutter and stripper
- c. Electric tape / Insulation materials
- d. Hex/Allen key set
- e. Vernier Callipers

### **C. Motor System Used and Assembly**

- Foxeer Datura 1404 3850KV brushless motors configured in a quadcopter “X” layout
- 4× SimonK 30A ESCs individually mounted and thermally isolated to prevent overheating
- ESC–Motor soldered connections using high-temperature silicon wire
- Balanced GEMFAN 3016 propellers installed with thread-locked motor screws
- Power routed from LiPo battery through a distribution harness feeding the ESCs
- Signal lines from the ESCs connected to the ESP32-S3 PWM outputs for precise motor control



# Listing of all the Software used in Robo-making

## A. Software Used

- **Arduino IDE** – Used for firmware development, compilation, and flashing onto the ESP32-S3 and ESP32-CAM modules.
- **AutoCAD** – Utilized for mechanical design, frame modeling, and CAD-based structural refinement.
- **EdgeImpulse Studio** – Used for data collection, training, and deployment of mine-detection models onto the ESP32-CAM.
- **Python**– Employed for dataset preprocessing, SLAM map visualization, and debugging analysis.
- **Serial Plotter** – Used for tuning PID loops, visualizing IMU response, and validating RC input timing.

## B. Software Developed

- **Swarm Interaction Grammar Engine** – A custom logic layer that interprets human gestures and voice inputs into actionable swarm-level commands such as Survey, Regroup, or Follow Path.
- **Decentralized Collaborative SLAM (C-SLAM)** – A memory-efficient SLAM pipeline designed specifically for ESP32-class processors, enabling low-resolution depth-based mapping and shared occupancy-grid fusion.
- **Flight Controller Firmware** – Complete flight-control stack including complementary-filter-based attitude estimation, cascaded PID loops, motor mixing logic, safety mechanisms, and Boids-inspired swarm coordination behavior.
- **ESP32-CAM Mine Detection Module** – Onboard inference engine integrating the EdgeImpulse-trained CNN model for real-time identification of potential mine signatures.
- **Communication Layer for Swarm Networking** – Lightweight ESP-NOW/mesh protocol extensions enabling multi-drone data exchange without a central controller.



# Deviation from Level-1 document

1. **ESC Configuration:** Used four individual 30A ESCs instead of a 4-in-1 ESC due to unavailability of the required JST-SH connector in market.
2. **Mine Detection Sensor Change:** ToF sensors removed as they cannot reliably detect thin 5 mm discs; ESP32-CAM with ML inference adopted instead.
3. **Motor Substitution:** Foxeer Datura 1404 3850KV motors used because the originally proposed T-Motor F1404 units were out of stock.

## Application of the robot prepared



Mine Field Navigation



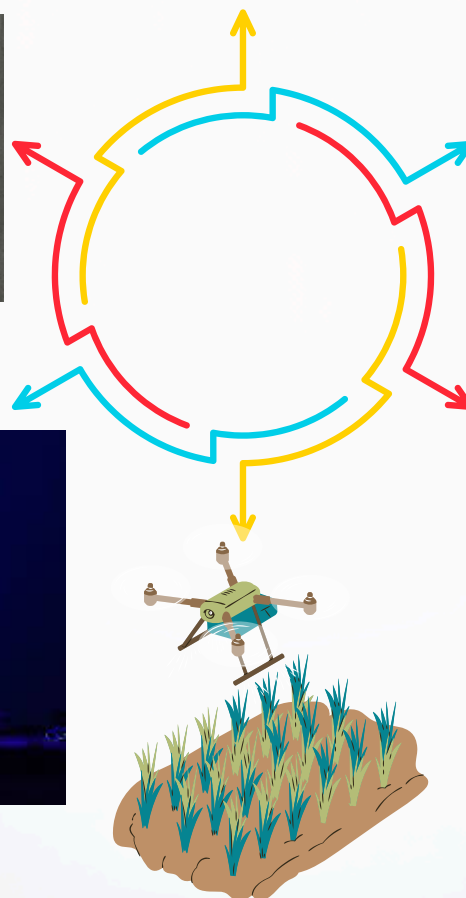
Infrastructure Management



Search and Rescue Operation



Drone Show

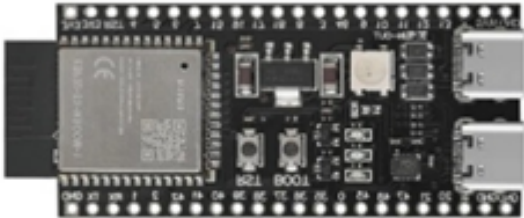
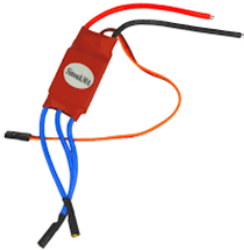


Agricultural Purposes

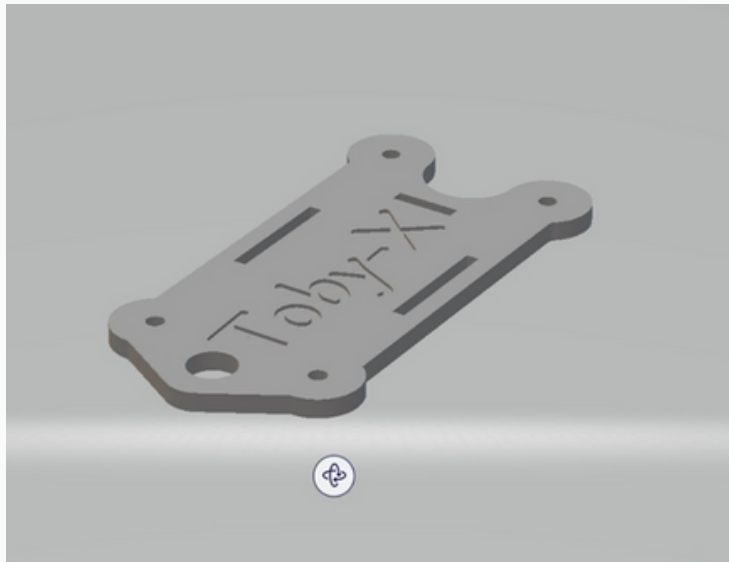


Crowd Management

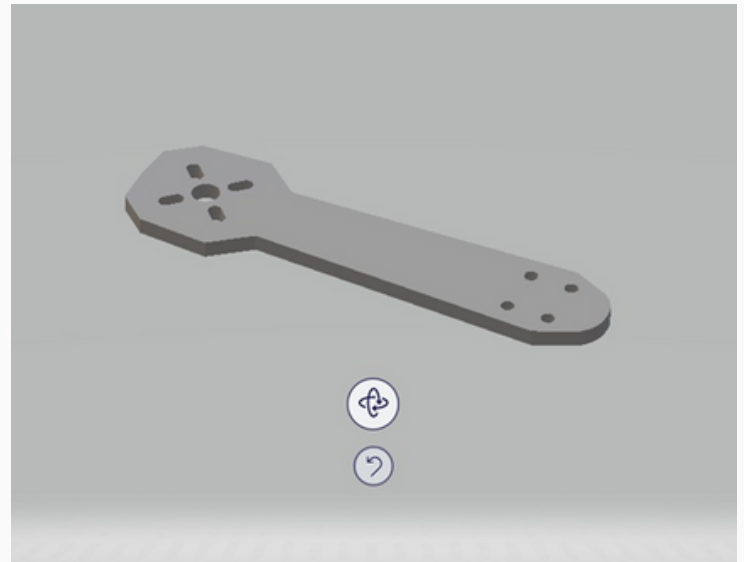


| Image of Component                                                                  | Name of Component         | Description                                                                                                                | Quantity |
|-------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------|----------|
|    | 3S 2200mAh LiPo           | High energy density and discharge rate for performance and >10 min endurance.                                              | 1        |
|    | Foxeer Datura 1404 3850KV | The 1404 motor required flight time for lightweight drones, avoiding strain and ensuring reliable coordination             | 4        |
|   | GEMFAN 3016 HURRICANE     | Offers efficient, low-noise flight with minimal vibration, precise control, and proven compatibility for 1404 motors on 4S | 4        |
|   | ESP32-S3-WROOM-1-N16R8    | Dual-core, vector instructions for AI, integrated Wi-Fi/BT, ESP-NOW support                                                | 1        |
|  | SimonK 30A ESC            | Controls brushless motors,                                                                                                 | 4        |
|  | MPU-6050                  | 6-DoF Accelerometer + Gyroscope. Standard for DIY drones, sufficient for stabilization.                                    | 1        |

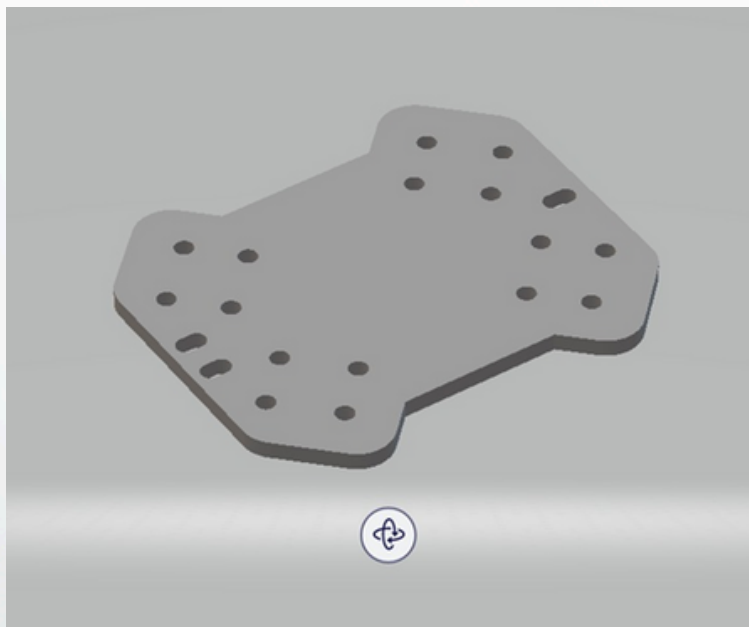




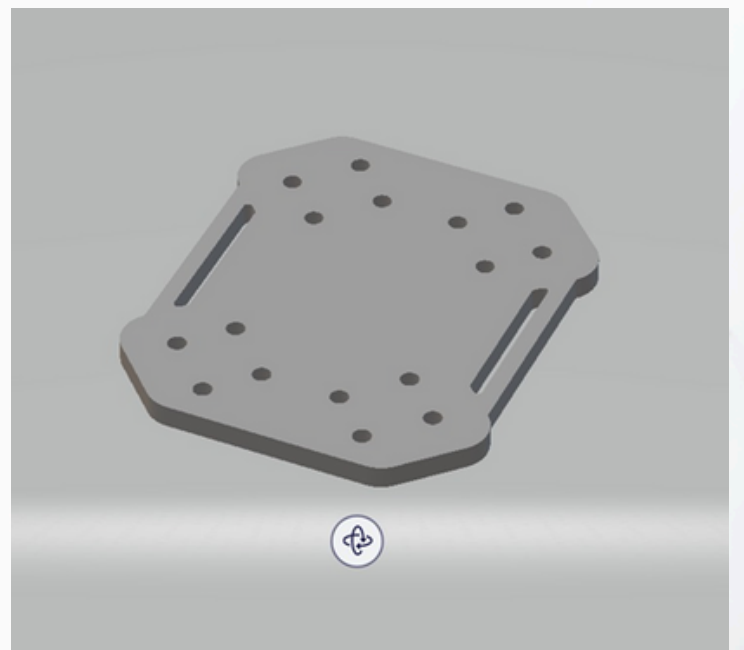
Cage Top



Arms



Base-1



Base-2



Before Assembly

**[Link to our Google Drive](#)**

**[Open it for more info and details of our working](#)**